

Experiment No: 6. Continuous Integration with Jenkins: Setting Up a CI Pipeline, Integrating Jenkins with Maven/Gradle, Running Automated Builds and Tests.

Step1: Create a new jenkins job

1. Log in to jenkins
2.
 - Open web browser and navigate to your Jenkins URL(<http://localhost:8080>)
 - Login with your admin credentials.
3. Install the required plugins
 - Go to" Manage Jenkins"
 - Select "Plugins (Available Plugins)"
 - Select: Maven Integration Plugin, Gradle plugin,Pipeline and Junit and click "Install"

Step 2: Select a tool For Maven:

1. Go to "Manage Jenkins"
2. Select "tools"
3. Go to" Maven Installations"
 - Click "add maven"
 - Uncheck the "Install automatically"
Provide "Name: Apache Maven 3.8.7"
"MAVEN_HOME:/usr/share/maven"
 - Save changes.

Step 3: Create a new job For Maven:

- On the Jenkins dashboard, click on "New Item"
- Enter an Item Name
- Select "Freestyle Project"
Click "OK"

Configure Source Code Management (SCM):In the job configuration page,

- Scroll down to “Build” and click on “Add build step”.
- Select “Invoke top-level Maven targets”.
- Goals :In the Goals field, enter: “**clean install**”
- Set the POM File location (</home/klecet/MavenProject/pom.xml>)

Save and build

Step 4: Select a Tool for Gradle:

1. Go to "Manage Jenkins"
2. Select "tools"
3. Go to "Gradle Installations"
 - Click "add gradle"
 - Uncheck the "Install automatically"
 - Provide "Name: **gradle8.4**"
"Gradle_HOME: **/opt/gradle/gradle-8.4**"
 - Save changes.

Step 5: Create a new job For Gradle:

- On the Jenkins dashboard, click on "New Item"
- Enter an Item Name
- Select "Freestyle Project"
Click "OK"

Configure Source Code Management (SCM)

- In the job configuration page, scroll down to the “advanced”
- Select "use custom workspace"-add directory: **/opt/gradle/gradle-8.4**
- Add Build Steps
- Scroll down to “Build” and click on “Add build step”.
- Select “Execute shell”.
- Enter the following into Command box
cd /home/klecet/gradle-projects
/opt/gradle/gradle-8.4/bin/gradle clean build
- Save and build

Step 6: Run Automated Tests in Jenkins

Configure Source Code Management (SCM)

- In the job configuration page, scroll down to the “advanced”
- Select "use custom workspace"-add directory: **/opt/gradle/gradle-8.4**
- Add Build Steps
- Scroll down to “Build” and click on “Add build step”.
- Select “Execute shell”.
- Enter the following into Command box
cd /home/klecet/gradle-projects
/opt/gradle/gradle-8.4/bin/gradle clean test
- Save and build

Step 7: Setting Up a CI Pipeline with Jenkins (Pipeline as Code)

Create a Pipeline Job

1. Log in to Jenkins and click on “New Item”.
2. Enter an Item Name: For example, Pipeline-CI.

Select “Pipeline” and click “OK”.

Define the Pipeline Script

1. Configure the Pipeline:

- In the job configuration page, scroll to the “Pipeline” section.
- Choose “Pipeline script”

2. Enter the Pipeline Script:(select from try sample pipeline)

```
pipeline {  
    agent any  
    stages {  
        stage ('Hello'){  
            steps {  
                echo 'Hello World'  
            }  
        }  
    }  
}
```

- Save and build.